MILITARY SPECIFICATION

TUBE, BLOOD COLLECTING, Vacuum, Sterile

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sterile vacuum blood collecting pilot tubes.

1.2 Classification. Sterile vacuum blood collecting pilot tubes shall be of the following types and sizes:

Type I - With citric acid-dextrose (AD solution) as additive
Type II - Without additive

Size 1 - 5.5 milliliters (ml.) nominal aspiration
Size 2 - 7 ml. nominal aspiration.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass (Laboratory).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

FSC 6630

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3.2 Construction. The blood collecting tubes shall consist of a tube and stopper and shall be in accordance with Figure 1. The stopper shall fit snugly in the open end of tube to form an airtight seal.

3.2.1 Tube. Tube shall be transparent and shall be fabricated of glass. Wall thickness shall be uniform. Bottom shall be round; open end shall be straight without a bead or flare.

3.2.2 Stopper. Stopper shall be fabricated of rubber and shall be a diaphragm molded in one piece. Top of diaphragm shall be flat with a recessed center and bottom shall be a hollow plug. Color shall be yellow cream. Coloring material shall be inorganic and non-bleeding. Shore durometer reading of stopper shall be 50 ± 5 when tested as specified in 4.4.1.

3.3 Additives. Type I tube shall contain 0.17 ml., +15% or -10%, AD solution. AD solution shall contain the following ingredients in the specified proportions:

Citric Acid, Hydrous, U.S.P.	1.52 grams (gm.)
Dextrose (Monohydrate), U.S.P.	5.0 gm.
Purified Water, qs	100.0 ml.

3.4 Sterilization assistant. Type II tube may contain 0.05 ml. of isotonic sodium chloride solution to produce steam for sterilization. Sodium chloride shall be reagent grade meeting the standards of Rosin's Reagent Chemicals and Standards, and water shall be distilled.

3.5 Performance.

3.5.1 Vacuum. When tested as specified in 4.4.2, minimum volumes of water drawn into tubes shall be as follows:

Size 1	5.0 ml.
Size 2	6.0 ml.

3.5.2 Resistance of tubes to centrifugal force. Resistance of tubes to centrifugal force shall be such that no breakage results when the tubes are tested as specified in 4.4.3.

3.6 Date of delivery. Date of delivery shall not be more than 60 days after the date of manufacture.

3.7 Expiration date. The expiration date shall be not less than 24 months from the date of manufacture. At the expiration date, the blood collecting tube shall draw at least 90% of the volume specified in 3.5.1.

3.8 Sterility. The interior of the sealed tube shall be sterile when tested with a biological indicator as specified in 4.4.4.

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3.8.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification
b. National Stock Number
c. Contract or purchase order number
d. Sterility lot or control number
e. Date of sterilization
f. Quantity of product in sterilizer lot
g. Type of sterilization (gas, radiation, heat, steam, etc.)
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product)
i. Test organism(s) and population density used
j. Test medium, incubation temperature, incubation time and dates of tests
k. Number of test units
l. Statement that samples comply with sterility tests
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA, 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.8.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.9 Workmanship. Workmanship shall be first class throughout. Tubes shall be free of defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the glass and rubber, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the names of the contractors, the contract numbers, the names of the manufacturers or suppliers, the NSN's, the item identifications, the names of the materials, the lot numbers, the lot sizes, the sample sizes, the date of testing, the test methods, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105. The inspection levels and acceptability quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be the tube.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	0.65
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product for test purposes shall be the tube.

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Table II Sampling for tests

Characteristic	Unit of product	Requirement	Test procedure	Inspection level	AQL
Hardness	Stopper	3.2.2	4.4.1	S-2	*
Vacuum	Tube	3.5.1	4.4.2	S-2	1.5
Resistance to centrifugal force	Tube	3.5.2	4.4.3	S-2	1.5
Sterility	Tube	3.8	4.4.4	S-3	*

* Acceptance number shall be zero.

4.3 Examination. The tubes shall be examined for defects including but not limited to those listed in Table III.

Table III Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Tube not free of cracks or chips.
102	Stopper not secured in tube.
103	Tube not free of dirt or foreign matter.
104	Tube does not contain additive (except Type II).
Minor	
201	Tube not free of sharp edges.
202	Tube not free of bubbles.
203	Stopper not free of bloom.

4.3.1 Dimensional examination. The tubes shall be examined for defects in dimension. Any dimension not within tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Durometer hardness. Stopper shall be tested in accordance with Method 3021 of FED-STD-601.

4.4.2 Vacuum. A seamless steel tube, Brown and Sharpe Gage No. 20, or one of the needles listed below shall be used for testing. The seamless steel tube shall have a point at one end and shall have a wall thickness from 0.0055 to 0.0060 in.; it shall be 2-1/2 + 1/16 in. long. One end of the testing tube or the venipuncture end of the needle shall be placed in distilled water and the pointed end of the testing tube or the stopper end of the needle shall pierce the diaphragm.

NSN 6515-00-404-2240 NSN 6515-00-492-8712
NSN 6515-00-404-2260 NSN 6515-00-864-6788

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4.4.3 Resistance to centrifugal force. Tubes shall be filled to capacity with distilled water and suspended in a centrifuge which has an angle head of 50 degrees from the vertical and a radius of 1-3/4 in. from the center of rotation to the center of the mouth of the tube shield. The centrifuge shall be accelerated gradually to a speed of 3000 revolutions per minute (r.p.m.). Centrifuge shall run at 3000 r.p.m. for 30 minutes. The current shall then be shut off and the centrifuge shall be allowed to stop without the use of a brake.

4.4.4 Sterility test. The interior of the tube shall be tested for sterility in accordance with the method specified in the U.S.P. for Procedural Details for Sterility Tests, Class B, Type I, Inoculated Product or Inoculated Carrier for Non-Parenteral Devices, using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated in 4.4.4.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II.

4.4.4.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C.
Dry heat	B. stearothermophilus or B. subtilis var.niger (or globigii)	55° - 65°C. 30° - 35°C.
Gas	B. subtilis var. niger (or globigii)	30° - 35°C.
Radiation	B. pumilus	30° - 35°C.

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips), if used, shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.4 and 4.4.4.1. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.4.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests

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shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the contracting officer, and approval obtained before employment of the alternate method on a government contract.

4.4.4.4 Product testing. Testing for sterility of the product after receipt by the Government shall be without a biological indicator and in accordance with the U.S.P.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package containing 100 tubes, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging quantities

Types	Unit package	Intermediate package	Exterior container
I and II	1 unit	Not required	10 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a corrugated fiberboard box of appropriate size and design fabricated of 200 pound test fiberboard. Closure of box shall be adequate to prevent spilling of contents under normal handling. As an alternate, tubes may be packaged in a carton of appropriate size and design having a bursting strength of 125 pounds with polyfoam trays designed to hold tubes in individual cutout cells. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed

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for a type 2 load and constructed in accordance with PPP-B--585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L--10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproof in accordance with 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to Method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked in accordance with MIL-STD-129. Lot control number, contract or purchase order number, name of contractor and the following legend shall be shown: "STORE IN A COOL PLACE (59°F. - 86°F.)". Type II shelf life markings as specified in 5.2.2.1 (b) of MIL-STD-129 shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging. (Example: "A-9/69-G".) NOTE: Final sterilization shall be on or after indicated packaging date. For Type II, a printed sheet with the following statement shall be included in each unit package: "MAY CONTAIN DROPLETS OF ISOTONIC SALT SOLUTION WHICH WILL NOT INTERFERE WITH SEROLOGICAL TESTS. NOT TO BE USED FOR BLOOD CHEMICAL ANALYSES".

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Sterility lot control number and the following legend shall be shown: "STORE IN A COOL PLACE (59°F. - 86°F.)". Type II shelf life markings as specified in 5.5.1 (b) of MIL-STD-129 shall be shown. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry heat sterilization) shall be shown following the level of packaging and date of packaging. (Example: "A-9/69-G".) NOTE: Final sterilization shall be on or after indicated packaging date.

NOTE: The lot control number shall be traceable to each sterilized charge.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type, size and grade required (see 1.2).
c. National Stock No.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 The items covered by this specification are for use as pilot tubes with the following items listed in Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-926-4766	Blood Collecting-Dispensing Bag and Donor Set
6515-00-926-4767	Blood Collecting-Dispensing Bag and Donor Set, Double Bag
6515-00-404-2240	Needle, Hypodermic, Blood Collecting, Double Pointed, 20 Gage, 1 inch, 12s
6515-00-492-8712	Needle, Hypodermic, Blood Collecting, Double Pointed, Disposable, 20 Gage, 1-1/2 inch with shut-off sleeve, 100s
6515-00-864-6788	Needle, Hypodermic, Blood Collecting, Double Pointed, Disposable, 20 Gage, 1-1/2 inch, 100s
6630-00-404-2220	Holder, Blood Collecting Tube, 12s
6515-00-404-2260	Needle, Hypodermic, Blood Collecting, Double Pointed, 20 Gage, 1-1/2 inch, 12s

6.3 Tubes with stoppers fabricated from the specified rubber have been found suitable for collecting, containing and transporting blood samples. Stoppers fabricated from other kinds of rubber will be considered if detailed reports on the following are submitted: durability of the seal and effect of the rubber on the results of the applicable blood analysis.

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

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Navy - MS
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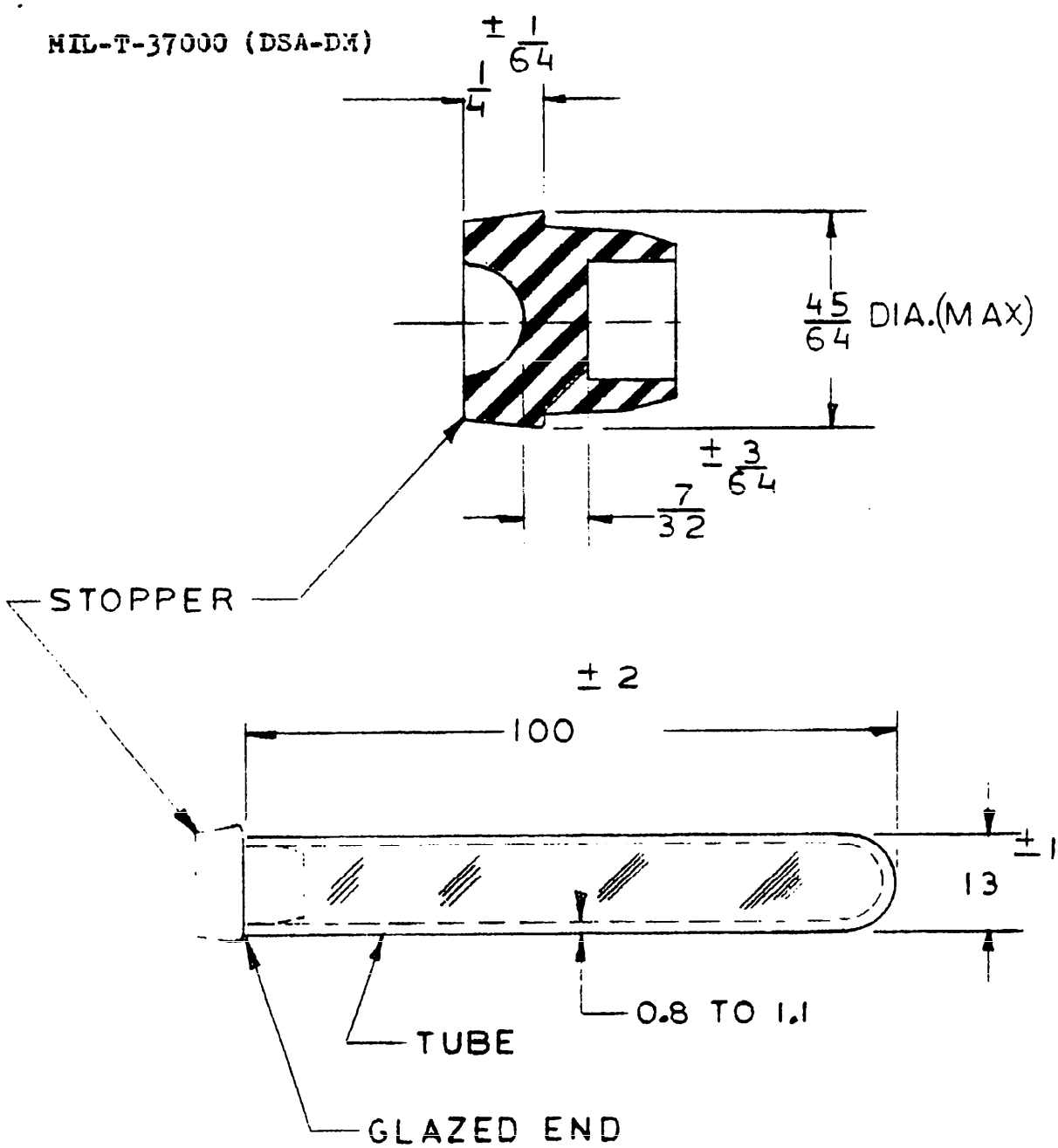
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0220

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STOPPER DIMENSIONS ARE IN INCHES
TUBE DIMENSIONS ARE IN MILLIMETERS

FIGURE-1- TUBE AND STOPPER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

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MIL-T-37000 (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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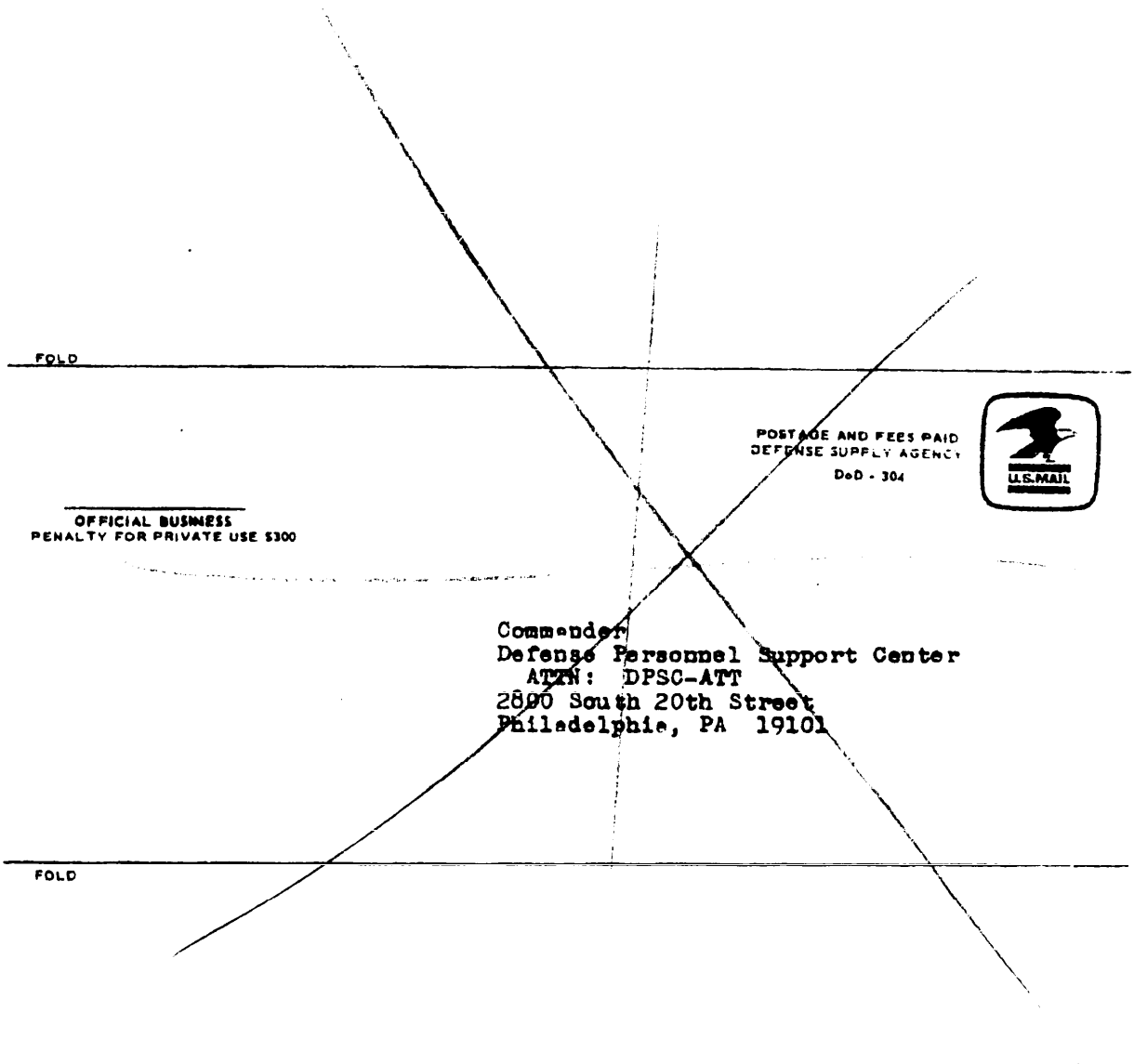
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